

ESTIMATING PARAMETERS IN AN AUTOREGRESSIVE PROCESS BY COMBINING THE METHODS OF MOMENTS AND OF LEAST SQUARES

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Key words: Method of moments, method of generalized least squares, gamma distribution, first order autoregressive process.

SUMMARY

A generalized least squares estimation procedure using more than one sample statistic is discussed in this paper. This method is illustrated by estimating the parameter of a standard gamma distribution and is then used to derive new estimators for the parameter of a first order autoregressive process. A comparison by means of simulation studies is made between some of these estimators and r_1 , the first order sample autocorrelation coefficient, which is the usual estimator for the parameter of this process.

1. Introduction

The method of moments for estimating population parameters from a sample is well-known and has been used continuously over the last century. If two or more moments are calculated from the same sample these moments are of course correlated. The classical method of moments does not take this correlation into account.

An adjusted method of moments using the correlation between moments does exist but is rarely mentioned in text books. This method which we shall now consider further is a combination of the method of moments and of generalized least squares. Similar methods have been employed by: Hinz and Gurland, (1967); Mather and Jinks, (1971, p. 156-164); Browne, (1974), (1977); Christofferson, (1975); Steyn, (1976); Muthen, (1978).

The aim of this paper is:

- (i) to illustrate and assess this adjusted method of moments by estimating the parameter of a standard gamma distribution, using the first four sample moments and
- (ii) to derive new estimators for the parameter of a first order autoregressive process, using the first four sample autocorrelations.

A brief outline of the procedure is the following: Consider the estimation of p parameters $\beta_1, \beta_2, \dots, \beta_p$ from the statistics $S_1, S_2, \dots, S_t$ $t \geq p$, which are obtained from a sample of size n from a given population. Define the model

$$(1) \quad \underline{S} = \underline{f}(\underline{\beta}) + \underline{Y}$$

where the vectors $\underline{S} = (S_1, S_2, \dots, S_t)$ and $\underline{\beta} = (\beta_1, \beta_2, \dots, \beta_p)$. The vector $\underline{f}(\underline{\beta})$ consists of t functions of the β_i . Assuming that the mean of $\underline{Y}$ is asymptotically the zero vector and $\text{cov}(\underline{Y}) = \text{cov}(\underline{S}) = V$, we can define the "generalized least squares" estimator $\underline{b}$ of $\underline{\beta}$ as the vector which will minimize the quadratic form $\underline{Y}' V^{-1} \underline{Y}$ where $\underline{\hat{Y}} = \underline{S} - \underline{f}(\underline{b})$

When V is unknown, which is usually the case, it is replaced by a positive definite estimate $\hat{V}$.

In the case where all the elements of $\underline{f}$ are linear combinations of the elements of $\underline{\beta}$, the model (1) is linear, say,

$$(2) \quad \underline{S} = A \underline{\beta} + \underline{Y}$$

where it is assumed that A is a $t \times p$ matrix of known constants with full column rank. In this case minimization of $\underline{\hat{Y}}' \hat{V}^{-1} \underline{\hat{Y}}$ leads to

$$(3) \quad \underline{b} = (A' \hat{V}^{-1} A)^{-1} A' \hat{V}^{-1} \underline{S}$$

If we assume that $\hat{V}$ is consistent for V , i.e. $\text{plim}(n\hat{V}) = nV$, the following properties of $\underline{b}$ as defined in (3) can be derived:

$$(i) \quad \begin{aligned} \text{plim}(\underline{b}) &= (A' \text{plim}(n\hat{V})^{-1} A)^{-1} A' \text{plim}(n\hat{V})^{-1} \text{plim}(\underline{S}) \\ &= \underline{\beta} \text{ iff. } \text{plim}(\underline{S}) = A \underline{\beta} \text{ (which was assumed in (2)).} \end{aligned}$$

(ii) Using the Taylor expansion of $\underline{b} = \underline{h}(\underline{S})$ around $\underline{\beta}$ it follows that the asymptotic covariance matrix of $\underline{b}$ is equal to $G'VG$ with

$$G = \{\partial \underline{b} / \partial \underline{S}\} = \hat{V}^{-1} A (A' \hat{V}^{-1} A)^{-1}$$

$$\text{Therefore } G'VG = (A' \hat{V}^{-1} A)^{-1} A' \hat{V}^{-1} V \hat{V}^{-1} A (A' \hat{V}^{-1} A)^{-1}$$

which is asymptotically equal to $(A' V^{-1} A)^{-1}$.

(iii) When $S_1, S_2, \dots, S_t$ are asymptotically normally distributed (e.g. when they are sample moments) then it follows that the limiting distribution of $\underline{\hat{Y}}' \hat{V}^{-1} \underline{\hat{Y}}$ is the chi-square distribution with $(t-p)$ degrees of freedom. This result provides a test of fit of the model (2).

2. Estimation of the parameter of the standard gamma distribution

The maximum likelihood estimator of β , the mean of the standard gamma distribution with density

$$(4) \quad f(x) = (1/\Gamma(\beta))x^{\beta-1}e^{-x}, x > 0, \beta > 0,$$

for a sample $x_1, x_2, \dots, x_n$, is the solution of the loglikelihood equation

$$(5) \quad -n(d/d\beta) \ln(\Gamma(\beta)) + \sum_{i=1}^n \ln(x_i) = 0$$

The function $(d/d\beta)\ln(\Gamma(\beta))$ is tabulated (e.g. M. Abramowitz and I.A. Stegun, (1964)) and therefore the maximum likelihood estimator can be approximated using equation (5). A less roundabout estimator which is often used is the sample mean m'_1 . To illustrate the adjusted method of moments as described in section 1, we shall now proceed to obtain estimators which are easily computable but preferable to m'_1 .

The reason for choosing the standard gamma distribution to illustrate and assess this adjusted method of moments lies in the fact that the first three moments μ'_1, μ'_2 and μ'_3 are linear functions of the unknown parameter.

The first four moments of X with density (4) are given by:

$$(6) \quad \mu'_1 = \beta, \mu'_2 = \beta, \mu'_3 = 2\beta, \mu'_4 = 3\beta(\beta+2)$$

In the notation of equation (2) we have, using the two moment equations $S_1 = m'_1 = \beta + Y_1$ and $S_2 = m'_2 = \beta + Y_2$, that

$\underline{S} = A\beta + \underline{Y}$ where $A' = (1, 1)$ and the covariance matrix of $\underline{S}$ is (omitting terms of $O(n^{-2})$)

$$(7) \quad V = (1/n) \begin{pmatrix} \mu_2 & \mu_3 \\ \mu_3 & \mu_4 - \mu_2^2 \end{pmatrix}$$

or by (6)

$$(8) \quad V = (1/n) \begin{pmatrix} \beta & 2\beta \\ 2\beta & 2\beta(\beta+3) \end{pmatrix}$$

V will be estimated from the moments of the sample. Two methods for estimated V are immediately suggested by (8) and (7) above viz.:

$$\hat{V} = (1/n) \begin{pmatrix} m'_1 & 2m'_1 \\ 2m'_1 & 2m'_1(m'_1+3) \end{pmatrix}$$

and by

$$V^* = (1/n) \begin{pmatrix} m_2 & m_3 \\ m_3 & m_4 - m_2^2 \end{pmatrix}$$

Substituting $\hat{V}$ and V^* for V it follows from (3) that β may be estimated respectively by

$$\hat{\beta} = [2(m'_1+2)m'_1 - m_2]/(2m'_1+3),$$

and

$$\beta^* = [(m_4 - m_2^2 - m_3)m'_1 + (m_2 - m_3)m_2]/(m_4 - m_2^2 + m_2 - 2m_3)$$

In order to examine the four estimators viz. m'_1 , $\hat{\beta}$, β^* and the maximum likelihood estimator β_{ml} we consider the relative efficiencies of the estimators. We define the asymptotic relative efficiency (a.r.e.) of an asymptotic unbiased estimator of β by the ratio of the Cramer-Rao lower bound and the asymptotic variance of the estimator (see e.g. Kendal and Stuart, Vol. 2, (1961), p. 9). The asymptotic variance of β_{ml} equals the Cramer-Rao lower bound which is the inverse of

$$\{-n \cdot E[\frac{d^2}{d\beta^2} \ln f(x_i)]\}$$

From (4) it follows that

$$\frac{d^2}{d\beta^2} \ln f(x_i) = -\frac{d^2}{d\beta^2} \ln \Gamma(\beta) = -\xi(2, \beta)$$

where $\xi(a, b) = \sum_{i=0}^{\infty} (b+i)^{-a}$ is known as Riemann's Zeta function (e.g. Gradsteyn and Rhyzik, (1965)).

Therefore

$$\text{var}(\beta_{ml}) \xrightarrow{n} [n \sum_{i=0}^{\infty} (\beta+i)^{-2}]^{-1}$$

The asymptotic variances of $\hat{\beta}$ and β^* are equal and follow from property (ii) section 1

as

$(A'V^{-1}A)^{-1}$ where $A' = (1,1)$ and V is defined in (8). Therefore

$$\text{var}(\hat{\beta}) \xrightarrow{n \rightarrow \infty} 2\beta(\beta+1)/n(2\beta+3)$$

Finally the variance of m'_1 is β/n .

Table 1 compares the a.r.e. of m'_1 , $\hat{\beta}$ and $\beta_{m\ell}$ for different values of β . This table shows that $\hat{\beta}$ and β^* are asymptotically preferable to m'_1 and almost as reliable as $\beta_{m\ell}$ if $\beta \geq 3$. Other estimators can be derived when the three moment equations $m'_1 = \beta + Y_1$, $m_2 = \beta + Y_2$ and $m_3 = 2\beta + Y_3$ are used.

3. Estimating the parameter of a first order autoregressive process

Consider the time series $\underline{u}' = (u_1, u_2, \dots, u_T)$ where

$$(9) \quad u_t - \mu = \rho(u_{t-1} - \mu) + \varepsilon_t, |\rho| < 1, t = 1, 2, \dots, T.$$

Let $\{\varepsilon_1, \varepsilon_2, \dots\}$ be a stationary stochastic process with

$$E(\varepsilon_t) = 0, E(\varepsilon_t^2) = \sigma^2 < \infty, t = 1, 2, \dots, T.$$

Assume finally that the variables $\varepsilon_1, \varepsilon_2, \dots$ are uncorrelated. The correlation matrix of this time series is

$$P = \begin{bmatrix} 1 & \rho & . & . & . & \rho^{T-1} \\ \rho & 1 & . & . & . & \rho^{T-2} \\ . & . & . & . & . & . \\ \rho^{T-1} & \rho^{T-2} & . & . & . & 1 \end{bmatrix}$$

Define the sample autocorrelation coefficients when μ is known (we can assume without loss of generality that $\mu = 0$) by

$$r_k = \frac{1}{T-k} \sum_{t=1}^{T-k} u_t u_{t+k} / \frac{1}{T} \sum_{t=1}^T u_t^2, k = 1, 2, \dots, T-1.$$

In the case of a time series with unknown mean μ the autocorrelations are defined as (e.g. Fishman, (1968), p. 83).

$$r_k^{(1)} = \frac{1}{T-k} \sum_{t=1}^{T-k} (u_t - \bar{u})(u_{t+k} - \bar{u}) / \frac{1}{T} \sum_{t=1}^T (u_t - \bar{u})^2,$$

$$k = 1, 2, \dots, T-1.$$

Under general conditions we have the following well-known results for large T and k , $j = 1, 2, \dots, T-1$.

$$(10) \quad \begin{aligned} E(r_k) &= \rho^k - 2k\rho^k/T + O(1/T^2) \\ E(r_k^{(1)}) &= \rho^k - \{(1+\rho)(1+\rho+\dots+\rho^{k-1}) + 2k\rho^k\}/T + O(1/T^2) \end{aligned}$$

(e.g. Marriot and Pope, (1954)).

$$(11) \quad \begin{aligned} \text{cov}(r_k, r_j) &= \text{cov}(r_k^{(1)}, r_j^{(1)}) \\ &= \frac{1}{T} \sum_{i=-\infty}^{\infty} (\rho_i \rho_{i+k-j} + \rho_{i+k} \rho_{i-j} - 2(\rho_i \rho_k \rho_{i+j} + \rho_i \rho_j \rho_{i+k} - \rho_k \rho_j \rho_i^2)) + O(1/T^2) \end{aligned}$$

where $\rho_s = \rho^{|s|}$. (e.g. Fishman, (1968), p. 86).

The least squares estimator of ρ in (9) when $\mu = 0$ is

$$\hat{\rho}_{LS} = \sum_{t=2}^T u_t u_{t-1} / \sum_{t=2}^T u_{t-1}^2$$

For T large and assuming that $\varepsilon_1, \varepsilon_2, \dots$ are independent normal variables the maximum likelihood estimator of ρ is

$$\hat{\rho}_{ML} = \sum_{t=2}^T u_t u_{t-1} / \sum_{t=2}^{T-1} u_t^2, \text{ (e.g. Anderson, (1971), p. 355)}$$

Hence in this case for T large we have $r_1 \doteq \hat{\rho}_{ML} \doteq \hat{\rho}_{LS}$. The same approximate relationship holds in the case of unknown μ (after changing the estimators accordingly).

In what follows in this section we shall assume without loss of generality that $\mu = 0$.

The question arises if it is advantageous to use higher order sample autocorrelations in the estimation of ρ . Linhart and Zucchini (1970), defined a least squares estimator $\hat{\rho}(t)$ for ρ by minimizing

$$(12) \quad \sum_{j=1}^t (\hat{\rho}^j - r_j)^2$$

by numerical methods.

Consider the model

$$\underline{r} = \underline{f} + \underline{Y}$$

which is of the form (1) with $\underline{g}' = \underline{f}' = (r_1, r_2, \dots, r_t)$ and $\underline{f}' = (\rho, \rho^2, \dots, \rho^t)$. Following the outline of section 1 we can obtain an estimate of ρ by minimizing

$$Q = (\underline{r} - \hat{\underline{f}})' V^{-1} (\underline{r} - \hat{\underline{f}})$$

where V is the covariance matrix of $\underline{r}$. As can be seen this is a weighted version of (12).

In observing or generating time series from process (9) with $\rho > 0$, it frequently happens that although the first two or three sample autocorrelations are positive, the higher order correlations fluctuate about zero and can therefore be negative. In view of this and in order to keep the estimation procedure simple we will in the rest of this paper confine our attention to the case of two correlation equations i.e. $t = 2$. However higher order autocorrelations could enter the estimation procedure by estimation of the unknown matrix V in terms of them.

The covariance matrix of $\underline{r}' = (r_1, r_2)$ can be derived from (11):

$$\text{var}(r_1) = \frac{1}{T} \sum_{i=-\infty}^{\infty} A_i + O(1/T^2)$$

$$\text{where } A_i = \rho^{|2i|} + \rho^{|i+1|+|i-1|} - 2(\rho^{1+|i|+|i+1|} - \rho^{2+|i|+|i+1|} - \rho^{2+|i|+|i-1|}).$$

Since it can be shown that $\sum_{i=-\infty}^{-1} A_i = -\sum_{i=1}^{\infty} A_i$, it follows that

$$\text{var}(r_1) = (1 - \rho^2)/T + O(1/T^2)$$

$$\text{Further } \text{cov}(r_1, r_2) = \frac{1}{T} \sum_{i=-\infty}^{\infty} B_i + O(1/T^2)$$

$$\text{where } B_i = \rho^{|i|+|i-1|} + \rho^{|i+1|+|i-2|} - 2(\rho^{1+|i|+|i+2|} + \rho^{2+|i|+|i+1|} - \rho^{3+|i|+|i-1|}).$$

Since it can readily be seen that $\sum_{i=-\infty}^{-2} B_i = -\sum_{i=2}^{\infty} B_i$, it follows that

$$\text{cov}(r_1, r_2) = \frac{1}{T} \sum_{i=-1}^1 B_i + O(1/T^2)$$

$$= 2\rho(1 - \rho^2)/T + O(1/T^2) \text{ after some simplification.}$$

In the same way it follows that

$$\text{var}(r_2) = (1 + 2\rho^2 - 3\rho^4)/T + O(1/T^2)$$

Ignoring terms of $O(1/T^2)$ the unknown matrix V is therefore

$$V = \frac{1}{T} \begin{pmatrix} 1 - \rho^2 & 2\rho - 2\rho^3 \\ 2\rho - 2\rho^3 & 1 + 2\rho^2 - 3\rho^4 \end{pmatrix}$$

There are different possibilities in the estimation of V :

(i) Estimate V from r_1 only:

$$\hat{V}_1 = [(1-r_1^2)/T] \begin{pmatrix} 1 & 2r_1 \\ 2r_1 & (1+3r_1^2) \end{pmatrix}$$

(ii) Estimate V from r_1 and r_2

$$\hat{V}_2 = [(1-r_1^2)/T] \begin{pmatrix} 1 & 2r_1 \\ 2r_1 & 1+3r_2^2 \end{pmatrix}$$

In both these cases the cubic equation $[dQ/d\rho]_{\rho=\hat{\rho}} = 0$ has the root $\hat{\rho} = r_1$ with the other two roots usually complex.

(iii) If we use r_1, r_2, r_3 and r_4 and estimate V by

$$\hat{V}_4 = \frac{1}{T} \begin{pmatrix} 1-r_2 & 2(r_1-r_3) \\ 2(r_1-r_3) & 1+2r_2-3r_4 \end{pmatrix}$$

it is interesting to note that $\hat{\rho} = r_1$ is not one of the roots of the equation $[dQ/d\rho]_{\rho=\hat{\rho}} = 0$. Simulation studies in this case revealed that the estimator obtained in the minimization of Q when V is estimated by $\hat{V}_4$, can be quite an improvement on r_1 . Unfortunately, the estimated value of ρ can also be erroneous. If ρ is restricted to the interval $|\rho| \leq 1$ in an iterative minimization procedure, there is a clustering of solutions at $\hat{\rho} = 1$ when the actual value of ρ is large.

In view of the importance of positive autocorrelation in time series, two alterna-

tive procedures which are based on the assumption that $r_2 \geq 0$ can be defined. These procedures are essentially the same as the minimization of Q , but avoid the problem of multiple roots.

$$\text{Consider } Q^* = (\underline{r}_* - \underline{v}\rho^*)' \hat{V}_*^{-1} (\underline{r}_* - \underline{v}\rho^*)$$

$$\text{with } \underline{r}'_* = (r_1, \sqrt{r_2}), r_2 \geq 0 \text{ and } \underline{v}' = (1, 1)$$

$\hat{V}_*$ being an estimator of V_* the covariance matrix for $\underline{r}_*$. Minimization of Q^* with respect to ρ^* leads to the generalized least squares estimator (which is a special case of (3))

$$(13) \quad \rho^* = (\underline{v}' \hat{V}_*^{-1} \underline{v})^{-1} \underline{v}' \hat{V}_*^{-1} \underline{r}_*.$$

Using the Taylor expansion or $\underline{r}_* = \underline{h}(r)$ around $\underline{v}\rho$ we have

$$V_* = J'VJ + O(1/T^2)$$

where J is the matrix of partial derivatives

$$J = \left\{ \frac{\partial \underline{r}_*}{\partial \underline{v}\rho} \right\}_{\underline{r}_* = \underline{v}\rho} = \begin{pmatrix} 1 & 0 \\ 0 & 1/2\rho \end{pmatrix}$$

Ignoring the terms of $O(1/T^2)$ it follows that

$$V_* = \frac{1}{T} \begin{pmatrix} 1-\rho^2 & (\rho-\rho^3)/\rho \\ (\rho-\rho^3)/\rho & (1+2\rho^2-3\rho^4)/4\rho^2 \end{pmatrix}$$

If V_* is estimated by means of r_1 only, or by both r_1 and r_2 , we have in both cases from (13) that $\rho^* = r_1$. If J and V are estimated separately from the first four autocorrelations, the estimator V_* can be formed

$$\hat{V}_* = \frac{1}{T} \begin{pmatrix} 1-r_2 & (r_1-r_3)/r_1 \\ (r_1-r_3)/r_1 & (1+2r_2-3r_4)/4r_2 \end{pmatrix},$$

Substituting this in (13) it follows that

$$(14) \quad \rho^* = (g_1 r_1 + g_2 \sqrt{r_2}) / (g_1 + g_2), r_2 \geq 0$$

with

$$g_1 = r_1 - 2r_1 r_2 - 3r_1 r_4 + 4r_2 r_3 \text{ and } g_2 = 4(r_2 r_3 - r_1 r_2^2).$$

As can be seen from (10) r_1 and $\sqrt{r_2}$ are biased estimators of ρ for $r_2 \geq 0$. It therefore can be argued that it is better to use the essentially unbiased estimators $r_1/(1-2/T)$ and $\sqrt{r_2/(1-4/T)}$ in the correlation equations. Another generalized least squares estimator of ρ , say $\tilde{\rho}$, can therefore be derived by minimization of

$$(15) \quad \tilde{Q} = (\underline{r}_* - \underline{w}\tilde{\rho})' \tilde{V}_*^{-1} (\underline{r}_* - \underline{w}\tilde{\rho})$$

with respect to $\tilde{\rho}$. Where $\underline{w}' = \{(1-2/T), \sqrt{1-4/T}\}$

This estimator reduces to

$$(16) \quad \tilde{\rho} = (\underline{w}' \tilde{V}_*^{-1} \underline{w})^{-1} \underline{w}' \tilde{V}_*^{-1} \underline{r}_*$$

In this case the matrix $\tilde{V}_*$ is derived by substituting ρ in V_* by the ordinary least squares estimator

$$\bar{r} = (\underline{w}' \underline{w})^{-1} \underline{w}' \underline{r}_*$$

which can be derived from the two correlation equations. Substituting $\underline{w}$ and $\underline{r}_*$ in $\bar{r}$ it follows that

$$\bar{r} = \{(1-2/T)r_1 + \sqrt{(1-4/T)r_2}\} / (2-8/T + 4/T^2)$$

Substituting $\bar{r}$ in V_* we have

$$\tilde{V}_* = \frac{1-\bar{r}}{T} \begin{pmatrix} 1 & 1 \\ 1 & (1+3\bar{r}^2)/4\bar{r}^2 \end{pmatrix}$$

Finally (16) reduces to

$$(17) \quad \tilde{\rho} = \frac{\{(1-2/T)A - \sqrt{1-4/T}\} r_1 + \{\sqrt{1-4/T} - (1-2/T)\} \sqrt{r_2}}{(1-2/T)^2 A - 2(1-2/T)\sqrt{1-4/T} + (1-4/T)}$$

where $A = (1+3r^{-2})/4r^2$

Since both $\hat{TV}_*$ and $\tilde{TV}_*$ are consistent estimators of TV_* , it follows from property (i) of section 1 that ρ^* and $\tilde{\rho}$ are both consistent estimators of ρ . From property (ii), it is evident that the variances of the three estimators r_1 , ρ^* and $\tilde{\rho}$ are asymptotically all equal to $(1-\rho^2)/T$. Since r_1 , r_2 are asymptotically normally distributed (Anderson and Walker, (1964)) it follows that r_1 and $\sqrt{r_2}$ are also asymptotically normally distributed for $r_2 \geq 0$. Therefore according to property (iii) of section 1 both Q^* and $\tilde{Q}$ are asymptotically distributed as chi-square with one degree of freedom. In the case of a time series (9) with unknown mean μ , the estimator ρ^* is still applicable by substituting $r_k^{(1)}$ for r_k , $k = 1, 2, 3, 4$. From (10) it follows that the approach which led to $\tilde{\rho}$ in the case of $\mu = 0$ would now involve nonlinear least squares and the estimator (16) can therefore not be applied.

4. Empirical comparison of the estimators

In order to compare r_1 , ρ^* and $\tilde{\rho}$ for small and intermediate sample sizes, sets of observations $(u_1, u_2, \dots, u_T)$ were generated according to model (9) with $\mu = 0$. In generating the observations it is easier to use the alternative form of (9).

$$(18) \quad u_t = \sum_{s=0}^{\infty} \rho^s \epsilon_{t-s}, \quad t = 1, 2, \dots, T$$

Since r_1 is approximately the maximum likelihood estimator of ρ when the $\{\epsilon_i\}$ are independent and identical normally distributed, it is asymptotically efficient in the normal case. Another error distribution was therefore used in the comparison of the three estimators. A convenient choice was to generate the ϵ_i in (18) as uncorrelated identical uniformly distributed random variables on the interval $(-\frac{1}{2}, \frac{1}{2})$.

In applying the estimators ρ^* and $\tilde{\rho}$ on the generated series, we are faced with some practical difficulties:

- (i) Although ρ is chosen to be positive, it can happen that $r_2 < 0$. In such cases ρ^* and $\tilde{\rho}$ are not defined.
- (ii) The absolute sum $|g_1 + g_2|$ defined in (14) can be small and this can lead to estimators $|\rho^*| > 1$.
- (iii) The matrix $\tilde{V}_*$ is always nonnegative definite but $\hat{V}_*$ can be indefinite for some series. This can also result in bad estimates ρ^* of ρ .

In view of this the following two dichotomous estimators will be defined

$$\tilde{\rho} = \begin{cases} \tilde{\rho} & \text{if } r_2 \geq 0 \\ r_1 & \text{if } r_2 < 0 \end{cases}$$

$$\rho^{**} = \begin{cases} r_1 & \text{if either } r_2 < 0 \text{ or } |g_1 + g_2| < a \text{ or } \det(\hat{V}_*) < 0 \\ \rho^* & \text{otherwise.} \end{cases}$$

The best choice of the constant a in the definition of ρ^{**} will depend on ρ and T . In comparing r_1 , $\tilde{\rho}$ and ρ^{**} a reasonable choice of $a = 0,1$ was made. The sample means, variances and mean squared errors for r_1 , $\tilde{\rho}$ and ρ^{**} for different combinations of T and ρ were calculated from 500 replications and can be seen in table II. n^{**} and $\tilde{n}$ respectively represent the number of cases (out of 500) where ρ^{**} and $\tilde{\rho}$ were the same or better estimators than r_1 while n_a is the number of times r_1 was chosen for ρ^{**} . From table II follows:

- (i) $\tilde{\rho}$ is essentially unbiased and r_1 has a larger bias than ρ^{**} .
- (ii) The sample variance of r_1 is in all but three situations the smallest and that of $\tilde{\rho}$ in all situations the largest.
- (iii) As far as mean squared error is concerned ρ^{**} is to be preferred to $\tilde{\rho}$ and seems also better than r_1 when ρ is larger than 0,7.
- (iv) Considering n^{**} and $\tilde{n}$ as criterion, the estimator ρ^{**} seems preferable to $\tilde{\rho}$.
- (v) The large variation in n_a shows that $a = 0,1$ is by no means an optimal choice and in general one can state that a should be inversely proportional to ρ .

Table III gives an indication of the applicability of the χ^2 -approximation of $\tilde{Q}$ and Q^* for small and intermediate sample sizes. Using only the cases where $\tilde{\rho}$ were chosen for $\tilde{\rho}_2$, the percentage of times that $\tilde{Q}$ (see (15)) exceeds the upper 10% and 5% values of the χ^2_1 distribution, were counted.

In the same way the last two columns give the percentage of times that Q^* exceeds these χ^2_1 values, counted only for these cases where ρ^* was chosen for ρ^{**} . From table III, it is clear that for all combinations of T and ρ considered, Q^* will lead to a very conservative test of fit while any test using $\tilde{Q}$ will have very large type I errors. However, the indications are that the approximation is already applicable for $\tilde{Q}$ in series where $T \geq 60$ and $\rho = 0,5$.

5. Autocorrelation in the linear model

The general linear model is usually formulated as

$$y = X\beta + u$$

with χ ($T \times 1$) the observed values on the dependent variable, X ($T \times K$) the observed values on the K regression variables and β ($K \times 1$) the unknown parameters. Assume that the errors $u_1, u_2, \dots, u_T$ are correlated and constitute a first order autoregressive process according to (9) with $\mu = 0$.

Define $\hat{u} = \chi - X\hat{\beta}$ with $\hat{\beta} = (X'X)^{-1} X' \chi$ the least squares estimator of β . Assuming that $\{\varepsilon_1, \varepsilon_2, \dots\}$ are independent and identical normally distributed, the Durbin-Watson test is usually used to test if $\rho = 0$. In the case of significant autocorrelation the Cochrane-Orcutt procedure is usually applied in order to improve the estimates $\hat{\beta}$. This procedure entails the following:

(i) Estimate ρ by $\hat{\rho} = r_1(\hat{u})$ the first order autocorrelation of the residuals $\hat{u}_1, \hat{u}_2, \dots, \hat{u}_T$.

(ii) Calculate $\hat{P} = \begin{bmatrix} 1 & \hat{\rho} & . & . & . & \hat{\rho}^{T-1} \\ \hat{\rho} & 1 & . & . & . & \hat{\rho}^{T-2} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ \hat{\rho}^{T-1} & \hat{\rho}^{T-2} & . & . & . & 1 \end{bmatrix}$

and estimate $\tilde{\beta}$ by means of the generalized least squares estimator

$$\tilde{\beta} = (X'\hat{P}^{-1}X)^{-1}X'\hat{P}^{-1}\chi$$

Using this value a new set of residuals can then be computed and ρ and $\tilde{\beta}$ can be re-estimated. This can be continued until convergence is obtained.

It is evident that $(\hat{u}_1, \hat{u}_2, \dots, \hat{u}_T)$ can only asymptotically be regarded as an observation of the stochastic process (9). In other cases the efficiency of the estimator $\hat{\rho}$ may be in doubt and a comparison of $\hat{\rho}$ and $\tilde{\beta}$ with alternative estimators seems reasonable. The results of section 4 suggest that ρ^{**} based on $r_1(\hat{u})$, $r_2(\hat{u})$, $r_3(\hat{u})$ and $r_4(\hat{u})$ could well be used as an alternative estimator to $\hat{\rho}$ in the case of significant positive autocorrelation. Although we are now only concerned with the initial estimation of ρ , the ρ^{**} estimator can be calculated and used in each iteration of the Cochrane-Orcutt procedure.

In order to compare these two estimators we considered the following simple model:

$$\begin{aligned} y_t &= 20 + 2.5t + u_t, t = 1, 2, \dots, T. \\ (19) \quad u_t &= \rho u_{t-1} + \varepsilon_t, t = 1, 2, \dots, T \end{aligned}$$

where $\{\varepsilon_1, \varepsilon_2, \dots\}$ are independent and identical normally distributed with mean zero and variance one. For the same values of ρ and T as in section 4 we simulated 100 time series $(y_1, y_2, \dots, y_T)$ according to (19). For each generated series the residuals $\hat{u}_1, \hat{u}_2,$

..., $\hat{u}_T$ were calculated and subsequently $\hat{\rho}$ and ρ^{**} . Table IV gives the results and all the symbols have the same meaning as in table II except that r_1 is replaced by $\hat{\rho}$. From this table the following are evident.

- (i) For both estimators we have a much larger bias in estimating ρ in model (19) than in model (18). This is expected, but the extent of the bias, especially for $T = 20$, is surprising. In all the cases considered, however, the bias of ρ^{**} is smaller than the bias of $\hat{\rho}$.
- (ii) Considering the mean squared error, the estimator ρ^{**} should be preferred for $\rho \geq 0,7$ and $T \geq 40$.
- (iii) The choice of $a = 0,1$ is by no means an optimal choice.

6. Conclusion

If the "moment equations" (1) is linear with respect to the unknown parameters, as in the case of a gamma distribution, the methods of estimation which are discussed in this paper, lead to estimators which are easy to calculate and which compare well with the usual estimators. The number of moment equations to be used as well as the best method of estimating V from a sample are two interesting theoretical problems for future consideration.

The estimators $\hat{\rho}$ and ρ^{**} for the parameter in a first order autoregressive process similar to the estimator r_1 do not make use of any a priori knowledge of the underlying distribution of the process $\{\epsilon_1, \epsilon_2, \dots\}$. When this distribution is known one should of course use the maximum likelihood estimator, e.g. Smit, (1963). In section 4 it was shown that the estimator ρ^{**} is to be preferred for large values of ρ and $T \geq 40$. The same conclusion can be made in applying this estimator in the general linear model with auto-correlated errors.

A disadvantage of the estimator ρ^{**} is the arbitrary choice of the constant a in it's definition. It will be difficult to determine optimal values of a for different values of (ρ, T) theoretically. This, however, can be done by means of simulation studies.

TABLE I
Comparison of the A.R.E. of $\hat{\beta}$ and m'_1
 β

Estimator	0,5	1	1,5	2	3	4	5	8	10	20
$\hat{\beta}$	0,540	0,761	0,856	0,905	0,950	0,970	0,981	0,994	0,997	1
m'_1	0,405	0,608	0,713	0,776	0,845	0,882	0,905	0,941	0,954	0,978

TABLE II
Sample statistics comparing estimators for 500 replications of series (18) with uniform errors
for various combinations of ρ and T

T	ρ	Mean value of			Variance of			Mean squared error of			Number out of 500		
		r_1	ρ^{**}	$\tilde{\rho}$	r_1	ρ^{**}	$\tilde{\rho}$	r_1	ρ^{**}	$\tilde{\rho}$	n^{**}	$\tilde{n}$	n_a
20	0,5	0,4431	0,4581	0,4843	0,0441	0,0464	0,0559	0,0473	0,0482	0,0561	337	277	241
	0,7	0,6351	0,6488	0,7048	0,0330	0,0349	0,0435	0,0372	0,0375	0,0435	352	255	200
	0,9	0,8080	0,8184	0,8896	0,0246	0,0237	0,0396	0,0331	0,0304	0,0397	432	262	273
40	0,5	0,4684	0,4799	0,4912	0,0206	0,0225	0,0234	0,0216	0,0229	0,0235	290	257	143
	0,7	0,6613	0,6729	0,6957	0,0166	0,0178	0,0186	0,0181	0,0185	0,0186	329	251	157
	0,9	0,8544	0,8603	0,8983	0,0093	0,0088	0,0106	0,0114	0,0104	0,0106	420	251	294
60	0,5	0,4833	0,4929	0,4990	0,0143	0,0155	0,0156	0,0146	0,0156	0,0156	277	247	100
	0,7	0,6732	0,6818	0,6965	0,0105	0,0109	0,0112	0,0112	0,0112	0,0112	324	255	140
	0,9	0,8676	0,8704	0,8976	0,0054	0,0053	0,0058	0,0064	0,0062	0,0058	423	255	329

TABLE III

Empirical level of significance for $\tilde{Q}$ and Q^* based on generated series (18) with uniform errors for small and intermediate sample sizes

T	ρ	$\tilde{Q} \geq \chi^2_{1,0,10}$	$\tilde{Q} \geq \chi^2_{1,0,05}$	$Q^* \geq \chi^2_{1,0,10}$	$Q^* \geq \chi^2_{1,0,05}$
20	0,5	18,13	11,73	4,25	1,54
	0,7	30,28	22,17	2,33	1,67
	0,9	41,90	34,01	0,88	0,88
40	0,5	12,73	6,36	5,04	2,24
	0,7	17,31	12,02	2,62	1,16
	0,9	41,00	34,00	1,46	0,49
60	0,5	10,70	7,64	4,75	3,5
	0,7	13,40	8,60	3,60	0,83
	0,9	33,80	27,20	0,58	0

TABLE IV

Sample statistics comparing various estimators for 100 replications of series (19)
for various combinations of ρ and T

T	ρ	Mean value of		Variance of		Mean squared error of		Number out of 100	
		$\hat{\rho}$	ρ^{**}	$\hat{\rho}$	ρ^{**}	$\hat{\rho}$	ρ^{**}	n_a	n^{**}
20	0,5	0,26780	0,27627	0,05320	0,05246	0,10712	0,10252	67	80
	0,7	0,40841	0,42132	0,04953	0,05733	0,13456	0,13499	49	86
	0,9	0,48640	0,49230	0,03767	0,04577	0,20873	0,21199	39	78
40	0,5	0,37846	0,38007	0,02126	0,02694	0,03603	0,04132	37	66
	0,7	0,57899	0,59068	0,01344	0,01605	0,02808	0,02800	18	62
	0,9	0,72710	0,74465	0,01322	0,01385	0,04311	0,03798	37	89
60	0,5	0,42350	0,43273	0,01693	0,01859	0,02278	0,02312	21	63
	0,7	0,61798	0,63146	0,01069	0,01132	0,01742	0,01602	13	63
	0,9	0,78080	0,78953	0,00941	0,00904	0,02362	0,02124	50	84

Acknowledgement

The author wishes to thank Professor M.W. Browne of the University of South Africa for helpful comments which lead to an improvement of this paper.

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Manuscript received 1979-04-11, revised 1980-05-13.